

Announcements

- Updated slides and code for Tue
- Assignment due today
- New assignment tomorrow

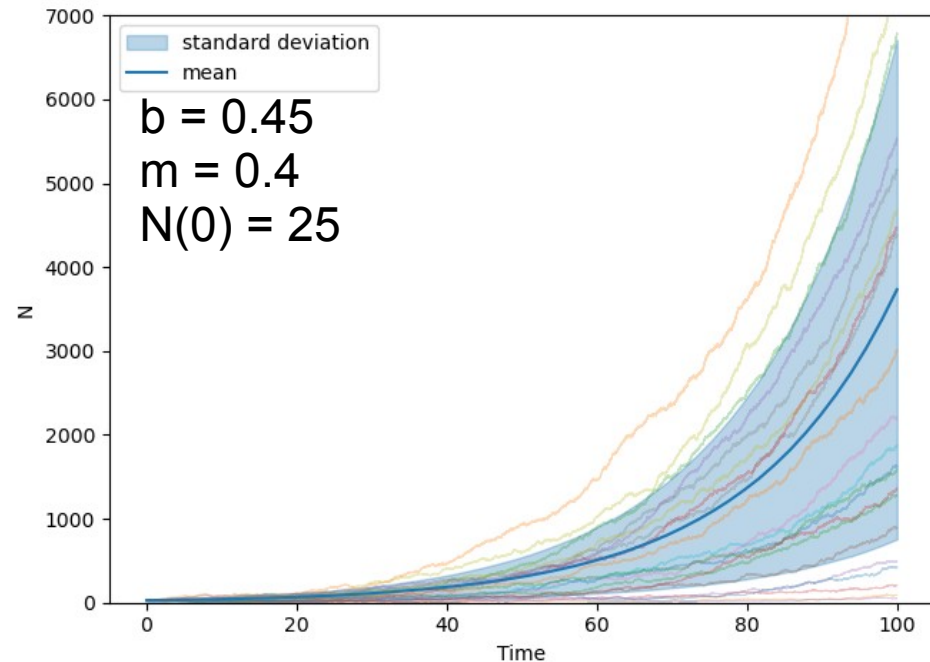
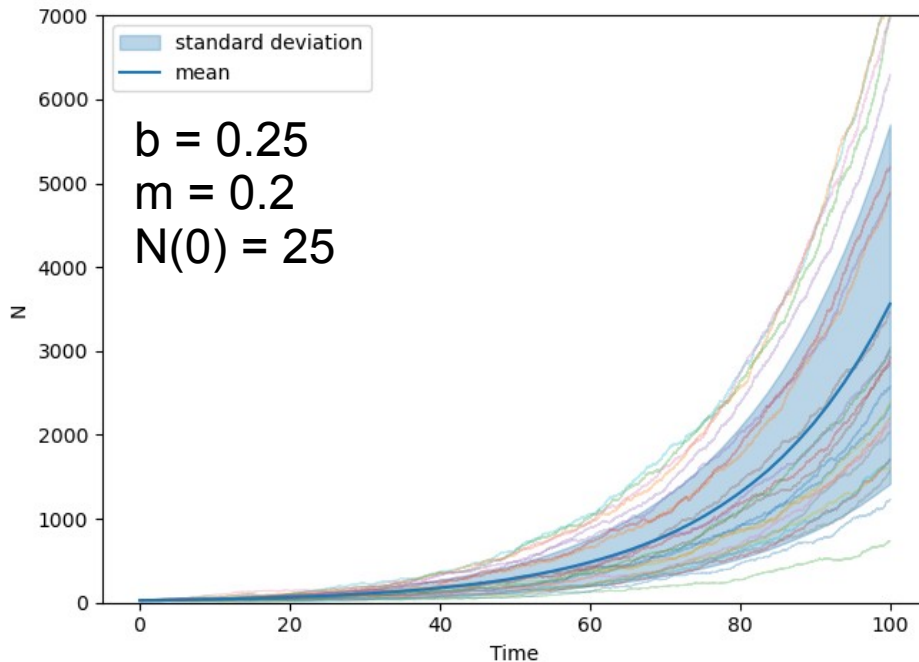
Today

- Deriving the deterministic model as the mean of the ensemble
- Deriving the variance of the ensemble
- Why variance matters
- Deterministic skeleton + noise emerges from mean + variance
 - diffusion approximation
- Variation among individuals

Why variance matters

Birth-death IBM: $\text{Var}(\Delta N) = (b + m) N(t) \Delta t$

Same growth rate ($r = 0.05$) $\rightarrow$ same mean dynamics



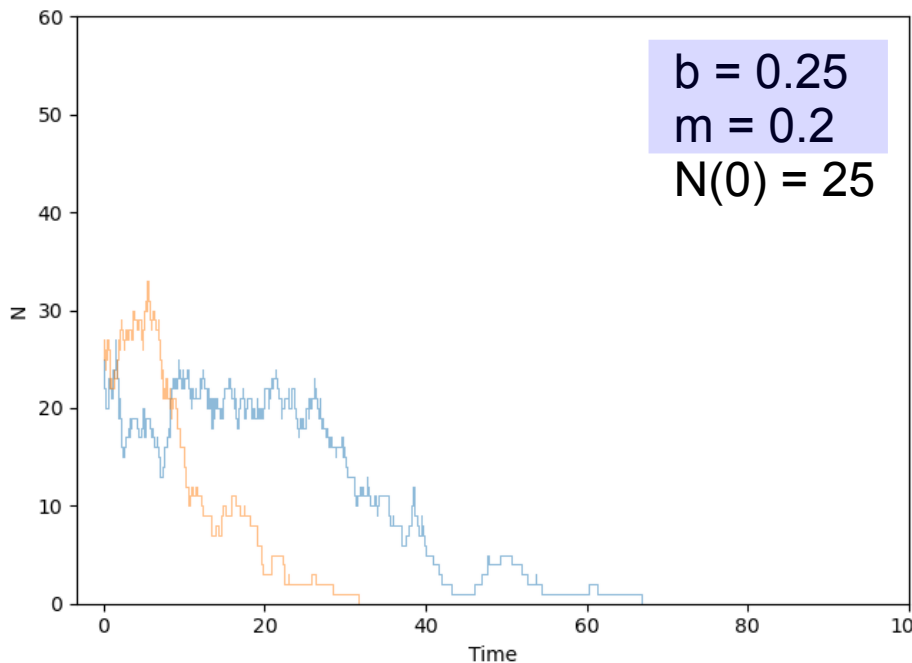
Same mean but **different variation** over time

sd = variation due to **demographic stochasticity**

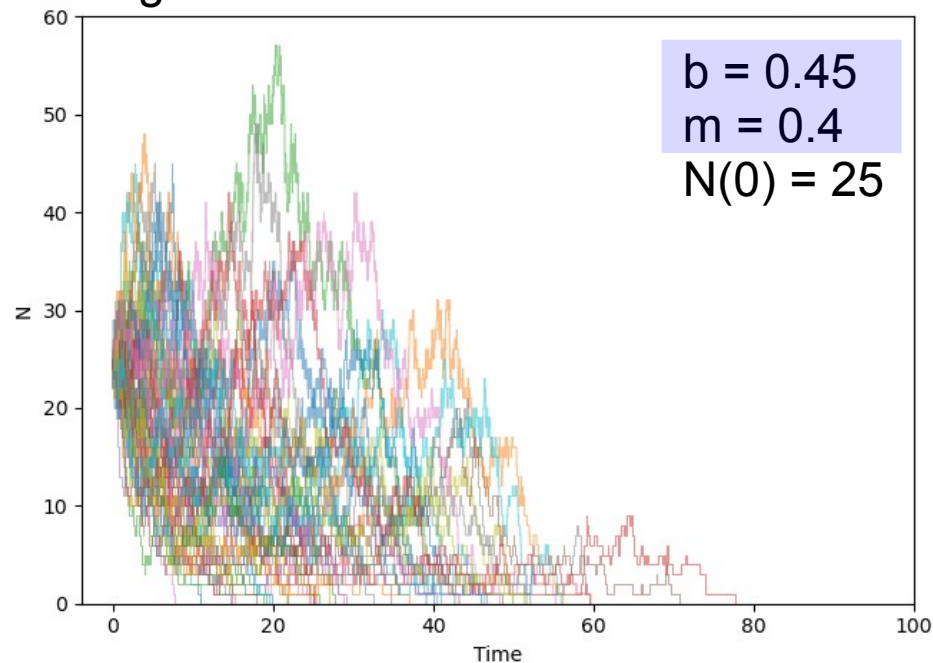
Why variance matters

Birth-death IBM: trajectories of **extinct** realizations

Lower variance: 2 extinct



Higher variance: 50 extinct



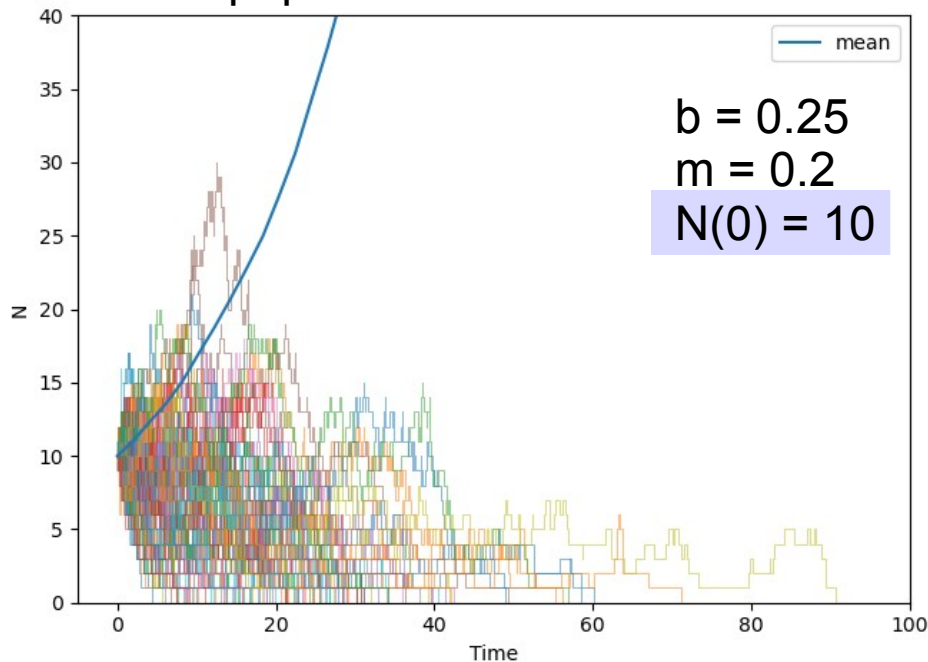
More extinction risk with higher demographic variance
due to **demographic stochasticity**

Population size matters too

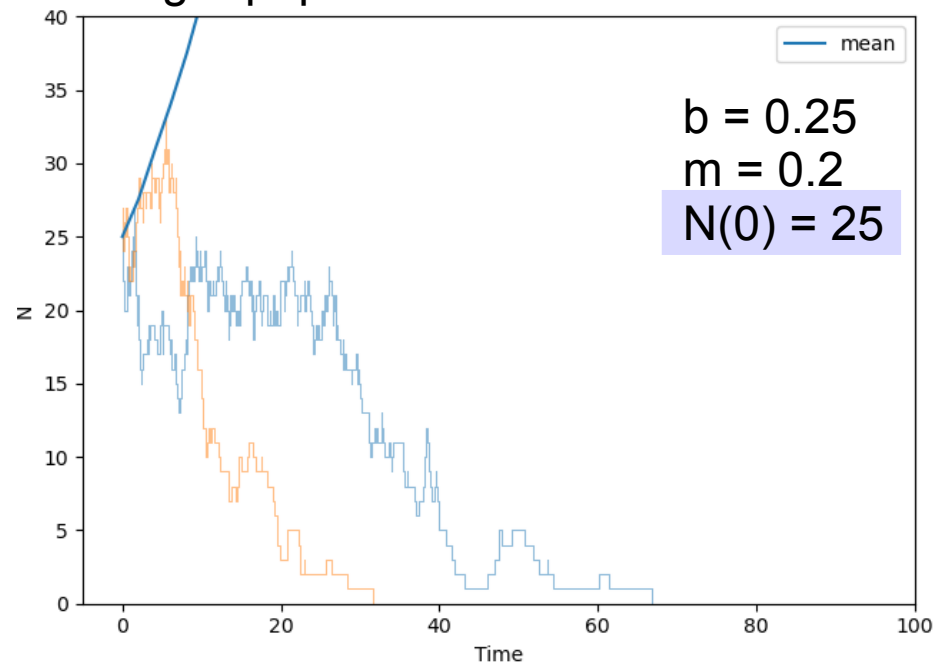
Birth-death IBM: trajectories of **extinct** realizations

Same variance, different population size

Small population: 114 extinct



Larger population: 2 extinct



More extinction risk with fewer individuals
demographic stochasticity matters most for **small populations**

Deterministic skeleton + noise

Deterministic skeleton function + stochastic function



Mean change and variance in change



Individual stochastic events

Deterministic skeleton + noise

We showed:

$$E[\Delta N] = (b - m) N(t) \Delta t$$

$$\text{Var}(\Delta N) = (b + m) N(t) \Delta t$$

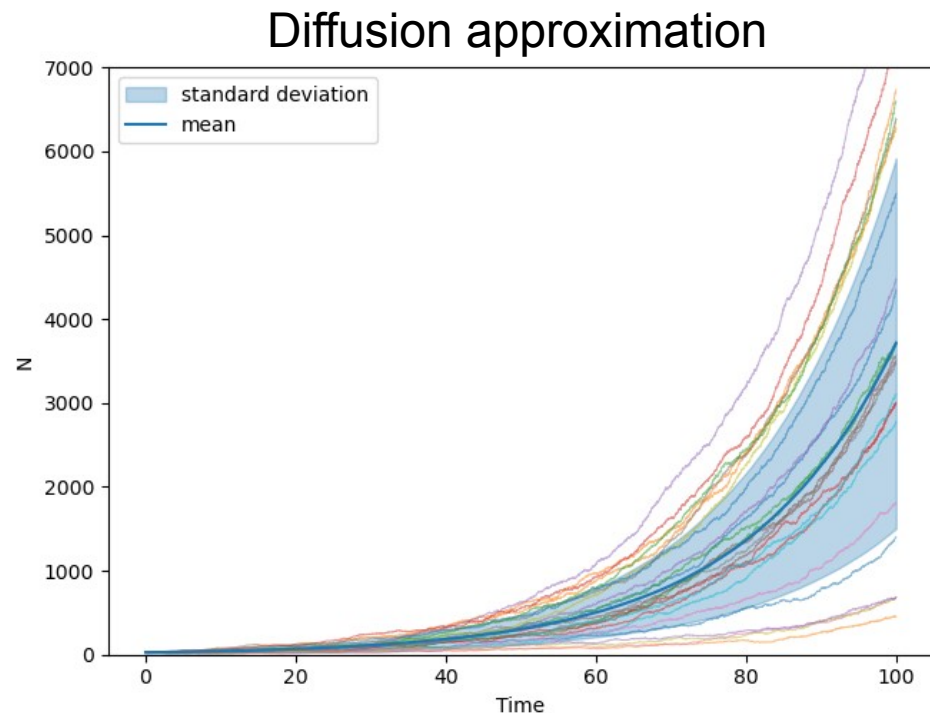
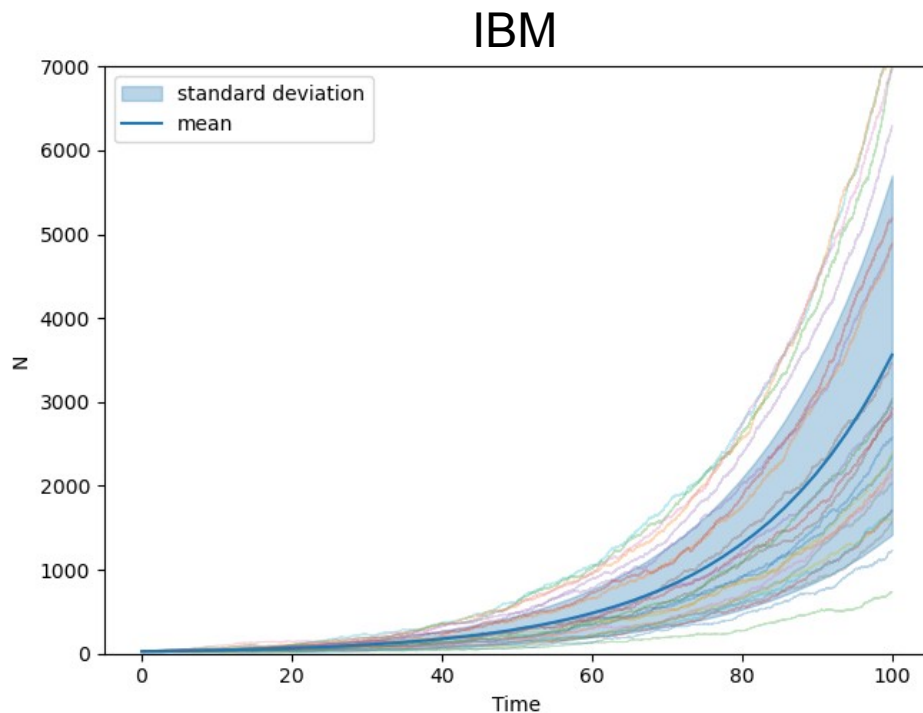
Suggesting **diffusion approximation**:

$$N(t + \Delta t) = \underbrace{N(t) + (b - m) N(t) \Delta t}_{\text{deterministic skel}} + \underbrace{\sqrt{(b + m) N(t) \Delta t} Z}_{\text{"noise"}}$$

$$Z \sim \text{Normal}(0, 1) \quad \leftarrow \text{CLT: because many small changes added together}$$

Deterministic skeleton + noise

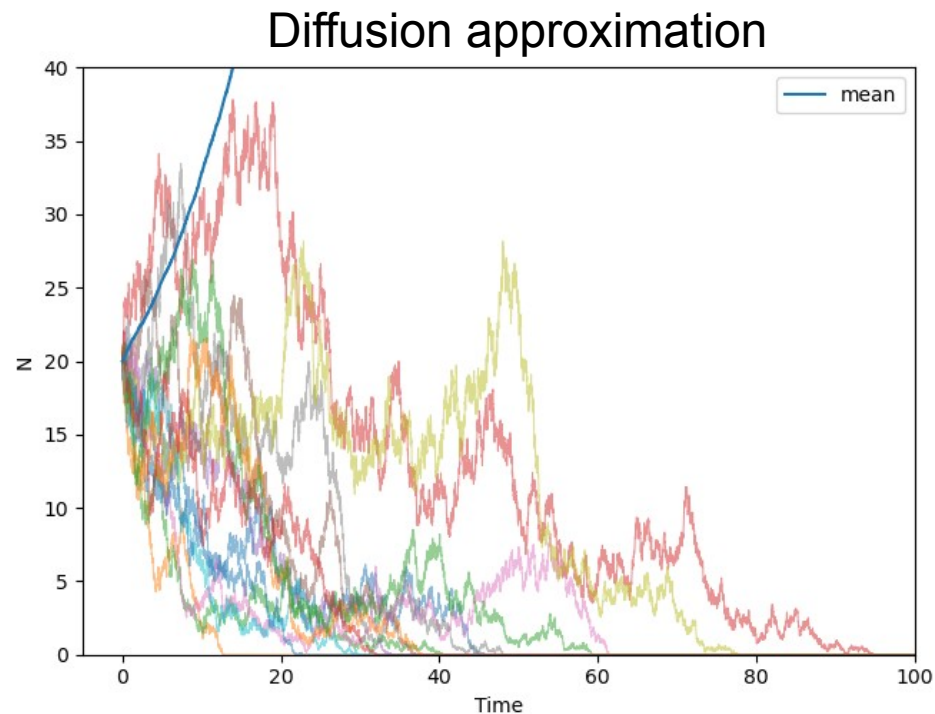
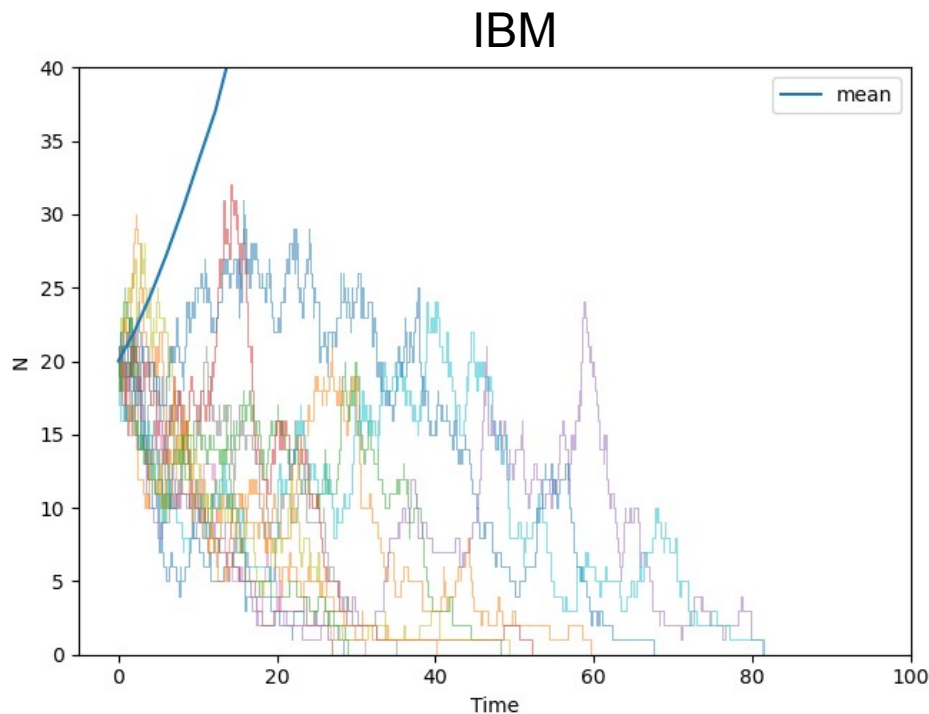
Same parameters: $N(0) = 25$, $b = 0.25$, $m = 0.2$



Diffusion approximation preserves mean and variance

Deterministic skeleton + noise

Trajectories of **extinct** realizations
Same parameters: $N(0) = 20$, $b = 0.25$, $m = 0.2$



Diffusion approximation preserves extinction dynamics

Heterogeneity among individuals

- Traits differ among individuals
 - e.g. birth rate, b , positive continuous variable
 - candidate distributions: gamma, lognormal

$$b_i \sim \text{Gamma}(k, \theta)$$

$$\bar{b} = k \theta$$

$$\text{Var}(b) = k \theta^2$$

$$\text{CV} = \frac{\sqrt{\text{Var}(b)}}{\bar{b}} = \frac{\sqrt{k}}{k}$$

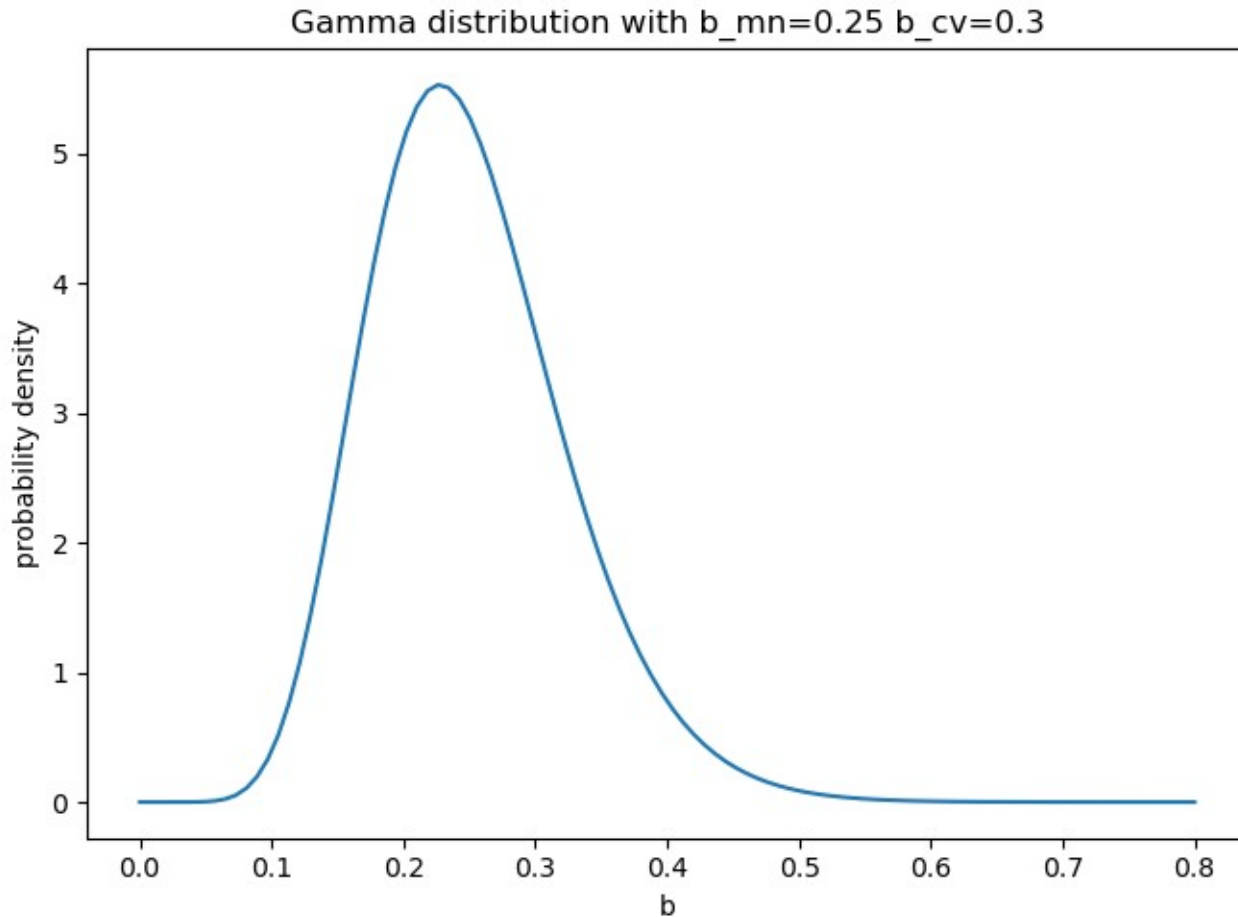
shape

$$k = \frac{1}{(\text{CV})^2}$$

scale

$$\theta = \bar{b} (\text{CV})^2$$

Heterogeneity among individuals



Code changes

IBM function

```
# b: vector of individual b values
# b_mn: mean birth rate, parameter of gamma distribution
# b_cv: variation in birth rate, parameter of gamma distribution

def birth_death_ibm_heterogeneous(alive, b, m, b_mn, b_cv, t_max, rng):    #<new

    # Initial biological state

    t = 0.0    #needs to be float, e.g 0.0

    # ---birth rate gamma distribution parameters
    b_shape = 1 / b_cv**2    #<new
    b_scale = b_mn * b_cv**2    #<
```

Code changes

IBM function

```
# which event type occurs?
event_birth = rng.binomial(1, b_intensity / intensity) #1 = birth

# which individual does the event happen to?
if event_birth:
    p_ind = b[alive_list[:N_t]] / b_intensity #<
    j = rng.choice(N_t, p=p_ind) #<
    i = alive_list[j] #<new
else: #<
    j = rng.integers(N_t) #<
    i = alive_list[j] #<
```

```
# update system state (do event)
if event_birth:
    # insert new individual in arrays
    alive[irow] = 1
    b[irow] = rng.gamma(b_shape, b_scale) #individual variation #<new
    m[irow] = m[i] #inherit from parent
    # update alive list
    alive_list[N_t] = irow
    # update intensities
    b_intensity = b_intensity + b[irow] #<new
    m_intensity = m_intensity + m[i]
```

Code changes

Simulation setup

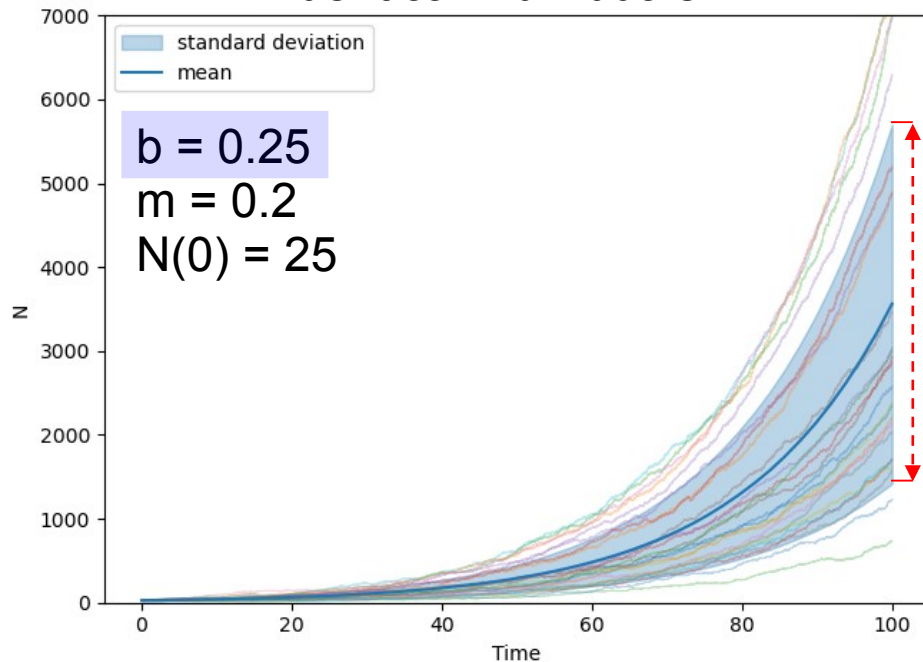
```
# Parameters and initial state
N_0 = 25
b_mn = 0.25
b_cv = 0.3
b_shape = 1 / b_cv**2
b_scale = b_mn * b_cv**2

alive = np.full(N_0, True) # alive state
b = rng.gamma(b_shape, b_scale, N_0) # birth rate
m = np.full(N_0, 0.2) # death rate
```

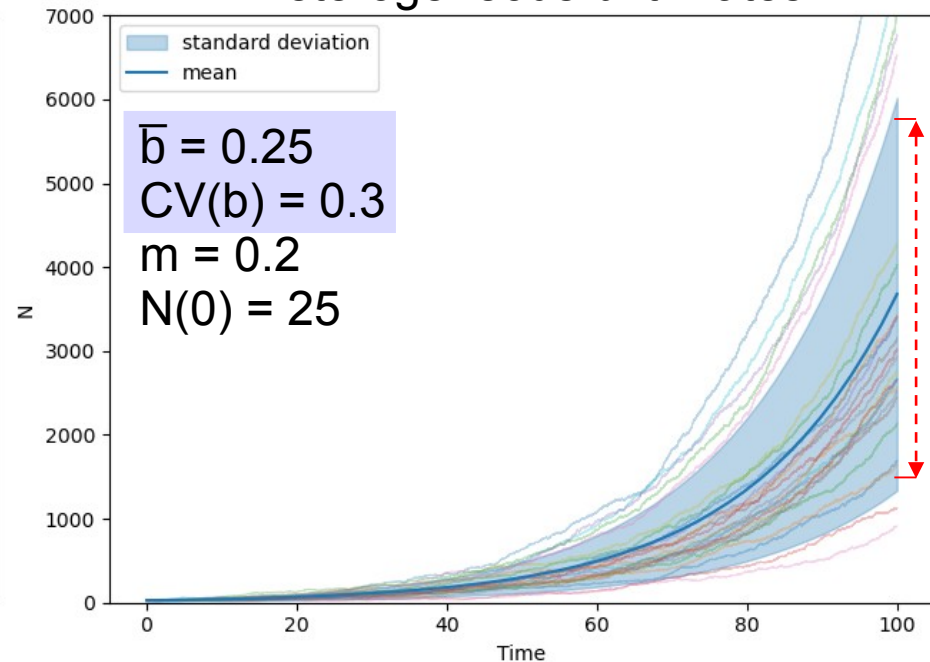
Heterogeneity among individuals

Birth-death IBM: $b_i \sim \text{Gamma}(\bar{b}, \text{CV}(b))$

Identical individuals



Heterogeneous birth rates



Same mean but slightly **larger variation**

sd = variation due to demographic stochasticity + **demographic heterogeneity**
extinction increased slightly (due to increased demographic variance)